

LIPID COMPOSITION GOVERNS MEMBRANE BINDING AND CONFORMATIONAL DYNAMICS OF THE MYOSIN-1C TAIL DOMAIN

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ABSTRACT

Class I myosins are essential linkers between the actin cytoskeleton and the plasma membrane, where they generate tension [1] and organise membrane nanodomains [2]. Myosin-1c (Myo1c) is a key example: its tail domain, composed of three calmodulin-bound IQ motifs, a post-IQ linker, and a pleckstrin homology (PH) domain, anchors the motor to the inner leaflet of the plasma membrane [3]. The PH domain exhibits selectivity for anionic phospholipids including PIP2 and phosphatidylserine (PS), yet the molecular basis for this specificity, and its consequences (if any) for the lever arm geometry and membrane orientation of the motor, remain unexplored. Here, we use experimental and molecular dynamics simulations to examine how lipid composition shapes the membrane-binding and conformational dynamics of the Myo1c tail domain in different biologically relevant membrane environments. Stable membrane association requires anionic lipids (PS and PIP2), as the protein rapidly dissociates from neutral DOPC membranes, whereas all anionic lipid conditions support stable membrane association. The PH domain tilt angle relative to the membrane normal is strongly lipid-dependent: DOPS yields the most upright, engaged orientation, whereas PIP2 alone results in a more tilted conformation, suggesting distinct binding modes for the two anionic lipid species. Analysis of the inter-segment bending angle at the IQ3/post-IQ hinge reveals that anionic lipid binding straightens this hinge region compared to the dissociated soluble condition, indicating that membrane engagement propagates conformational order from the PH domain anchor through the linker region toward the lever arm. Strikingly, combining PIP2 and DOPS partially disrupts the DOPS-bound geometry, suggesting competitive interplay between these two anionic lipid species in coordinating Myo1c membrane orientation. Residue-level binding site analysis identifies a conserved electrostatic hotspot within the PH domain that anchors both DOPS and PIP2. In the mixed PIP2/DOPS condition, providing a direct molecular explanation for the partial conformational disruption observed in the presence of both anionic lipid species. These orientation differences may have a consequence on the ability of the motor to associate with actin, since these in vitro experiments, actin association is strongly dependent on the lipid composition of the membrane bilayer. Taken together, these results provide a molecular picture of how inner-leaflet lipid composition controls both the affinity and geometry of Myo1c membrane binding, with implications for understanding how Class I myosins couple to cytoskeletal force generation at the inner membrane.

REFERENCES

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